

RAYYAN SHAN

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EDUCATION

University of Alberta

Bachelor of Science in Computing Science, Minor in Mathematics

Graduated April 2026

Edmonton, AB

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C

Frameworks: PyTorch, FastAPI, Django REST Framework, React, Vite

ML/Data: Computer Vision, Semantic Segmentation, U-Net++, OpenCV, NumPy, Pandas, scikit-learn, XGBoost

Tools: PostgreSQL, Docker, Linux, Nginx, Git, GitHub, Hetzner Cloud

EXPERIENCE

AI Research Software Developer

January 2026–Present

Quorum AI, University of Alberta

Edmonton, AB

- Built and deployed an AI-powered biomedical image-segmentation platform using PyTorch, U-Net++, FastAPI, React, TypeScript, and Docker.
- Developed the complete ML pipeline for 200 expert-annotated microscopy images, including preprocessing, augmentation, validation, inference, confidence estimation, and automated cell-metric generation.
- Created REST APIs and researcher workflows for image upload, segmentation visualization, TIFF export, and batch ZIP downloads; deployed to a Linux cloud server with Docker Compose and Nginx.
- Supported 2–10 oncology researchers processing 600+ images with substantially less manual segmentation; contributing as a co-author to the resulting manuscript.

Software Developer

June 2026–August 2026

Omintel

Remote

- Extracted and standardized 300+ enterprise AI use cases from technology-partner websites into a structured knowledge base.
- Designed a normalized taxonomy and search schema covering industries, business problems, AI capabilities, implementation approaches, and measurable outcomes.
- Evaluated LLM-assisted extraction workflows and collaborated with technical and business stakeholders to improve partner onboarding and AI-solution discovery.

PROJECTS

Cross-Session Neural Prediction Pipeline

Python, scikit-learn, XGBoost

- Built a leave-one-session-out machine-learning pipeline using CRCNS HC-6 neural-recording data and compared Ridge Regression, XGBoost, and nonlinear prediction models.
- Developed a residual-adaptation method that reduced error by 25–73% on difficult sessions and outperformed XGBoost on 8 of 9 evaluated subjects.

Cyan Federated Social Platform

Django REST Framework, React, PostgreSQL

- Developed a federated social platform with authentication, profiles, posts, comments, likes, follows, and server-to-server communication.
- Built REST APIs, automated tests, and backend services supporting distributed ActivityPub-style functionality.

PUBLICATIONS

AI-based Biomedical Image Segmentation for Automated Analysis of Fluorescence Microscopy Images

Co-author

Manuscript in preparation